Claims

Claim 1

A multi-agent interaction sincerity audit system based on logical continuity, characterized in that the system comprises:

1. A **causal logic continuity verification unit**: configured to extract a demand feature vector, a solution feature vector, a **physical action execution log hash** generated by a local secure zone, and a quantitative effect feedback feature vector during a multi-agent interaction session; the verification unit first calculates the **cosine similarity** between the demand feature vector and the solution feature vector to perform semantic alignment verification; when it is monitored that the solution is invoked and the corresponding quantitative effect feedback feature vector is generated, the verification unit uses the physical action execution log hash as a unique transaction index to retrieve the consensus deposition status of the distributed ledger, so as to verify whether the quantitative effect feedback feature vector satisfies a preset **causal self-consistency boundary** with the physical action execution log hash in terms of time sequence; when the matching degree of any layer is lower than a set threshold, or the corresponding physical action execution log hash is absent within a set time window, it is determined that a **Logical Disconnection** occurs;

2. A **privacy desensitization and signature verification unit**: connected to the causal logic continuity verification unit, deployed in a local sovereign sandbox, configured to perform in-situ causal determination on the quantitative effect feedback feature vector using a **hardware trusted isolated execution environment**, remove underlying sensitive plaintext including physical privacy parameters, generate desensitized Boolean determination proof data, and generate a cryptographic digital signature for the proof data using a hardware private key protected within the hardware sandbox, and send the signature to the distributed ledger;

3. A **Distributed Sincerity Audit Contract**: deployed in a distributed incremental right confirmation ledger, including a public key signature verification module, configured to verify the digital signature to confirm that it originates from the authorized hardware sandbox, and upon successful verification, use a sliding window model with a **decay factor** to update the **dynamic audit weight score** of the corresponding Logic Actor in an incremental iterative manner according to the current interaction consistency score contained in the proof data;

4. A **Three-Tier Gradient Disciplinary State Machine**: deployed within the Distributed Sincerity Audit Contract, configured to update the status mark of a Logic Actor to trusted, suspicious or fraudulent according to the dynamic audit weight score, and directly control the on-

chain settlement free quota of the Logic Actor according to the status category, as well as issue control instructions to the gateway layer to control the gateway communication handshake latency of the Logic Actor's commercial invocation requests by forcibly blocking the socket read/write buffer or injecting network-layer token bucket delay.

Claim 2

The multi-agent interaction sincerity audit system according to **Claim 1**, characterized in that:
The causal logic continuity verification unit determines interaction features of "claiming that the solution is invalid but no corresponding execution hash exists" or "claiming that the solution is invalid and the corresponding session request frequency shows exponential divergence" as **Logical Disconnection**, and directly issues a low sincerity rating result.

Claim 3

The multi-agent interaction sincerity audit system according to **Claim 1**, characterized in that:
The Distributed Sincerity Audit Contract updates the audit weight **W_t** using the following exponentially weighted moving average algorithm:

```
W_t = α · W_t-1 + (1 - α) · S_t
```

Where **α** is the decay factor and **0.8 ≤ α ≤ 0.95**, and **S_t** is the causal consistency score of the current interaction.

Claim 4

A multi-agent interaction sincerity audit method based on logical continuity, characterized in that, executed by a computer processor, the method comprises the following steps:
1. Four-dimensional logical chain comparison: during a multi-agent interaction session, extract a demand feature vector, a solution feature vector, a physical action execution log hash generated by a local secure zone, and a quantitative effect feedback feature vector; first calculate the cosine similarity between the demand feature vector and the solution feature vector to perform semantic

alignment verification; when it is monitored that the solution is invoked and the corresponding quantitative effect feedback feature vector is generated, use the physical action execution log hash as a unique transaction index to retrieve the consensus deposition status of the distributed ledger, so as to verify whether the quantitative effect feedback feature vector satisfies a preset causal self-consistency boundary with the physical action execution log hash in terms of time sequence; when the matching degree of any layer is lower than a set threshold, or the corresponding physical action execution log hash is absent within a set time window, determine that a Logical Disconnection occurs;

2. Local sandbox desensitization and on-chain recording: perform in-situ causal determination on the quantitative effect feedback feature vector within a local trusted isolation sandbox hardware enclave, remove underlying sensitive plaintext including physical privacy parameters, generate desensitized Boolean determination proof data, and generate a cryptographic digital signature for the proof data using a hardware private key protected within the hardware sandbox, and send the signature to the distributed ledger;

3. On-chain signature security verification: the Distributed Sincerity Audit Contract reads the signature and verifies the signature using the corresponding public key to confirm that it originates from the authorized hardware sandbox;

4. Dynamic audit weight evolution: upon successful signature verification, the contract uses a sliding window model with a decay factor to update the dynamic audit weight score of the corresponding Logic Actor in an incremental iterative manner according to the current interaction consistency score contained in the proof data;

5. Gradient level disciplinary control: update the trusted, suspicious or fraudulent status of the Logic Actor according to the dynamic audit weight score, directly control the on-chain settlement free quota of the Logic Actor according to the status category, and issue control instructions to the gateway layer to control the gateway communication handshake latency of the Logic Actor's commercial invocation requests by forcibly blocking the socket read/write buffer or injecting network-layer token bucket delay.

Claim 5

The multi-agent interaction sincerity audit method according to **Claim 4**, characterized in that:

In the four-dimensional logical chain comparison step, the similarity between the demand feature vector **V_D** and the solution feature vector **V_P** is calculated by the cosine angle formula:

```
Sim(V_D, V_P) = (V_D · V_P) / (||V_D|| ||V_P||) ≥ θ_consistency
```

When the obtained similarity is lower than the set value **θ_consistency**, a Logical Disconnection exception caused by demand-solution deviation is triggered; the temporal causality verification between the physical action execution log hash and the quantitative effect feedback feature vector includes: using the hash as a key of the decentralized database to retrieve the corresponding absolute timestamp of action occurrence, and calculating the difference between it and the capture timestamp of the effect feedback feature vector; if the difference exceeds the preset time window, or it is determined that the corresponding session request frequency shows exponential divergence while claiming no effect, a Logical Disconnection exception of solution-action or action-effect is triggered.

Claim 6

The multi-agent interaction sincerity audit method according to **Claim 4**, characterized in that:
In the local sandbox desensitization and on-chain recording step, the trusted isolated execution environment used is a hardware-level read-only protected enclave, which prevents external snooping and tampering on the audit logic or forgery and interception of proof data and signatures through physical memory isolation.

Claim 7

A ledger contract-side operation method in the multi-agent interaction sincerity audit method based on logical continuity, characterized in that, executed by an intelligent audit contract module deployed on a distributed ledger node, the method comprises the following steps:
1. Actor public key registration and binding: receive and save the public key binding transaction sent by a Logic Actor, map it to the ledger address of the Logic Actor, and establish an audit file;
2. Signature proof reception and verification: read the digital signature and desensitized audit determination data uploaded by the Logic Actor, perform one-click signature verification using the public key, and verify whether the signature originates from an authorized local hardware sandbox;
3. Sliding window weight update: for a determination data packet that passes signature verification, extract the causal consistency score contained therein, and substitute it into the sliding window with a decay factor to update the dynamic audit weight score of the corresponding actor;

4. Gradient state machine disciplinary response: when the obtained audit weight score drops to a set threshold, automatically freeze its revenue settlement funds, and issue control instructions to the gateway layer to control the gateway communication handshake latency of the Logic Actor's commercial invocation requests by forcibly blocking the socket read/write buffer or injecting network-layer token bucket delay.

Claim 8

A Logic Actor sandbox-side determination method in the multi-agent interaction sincerity audit method based on logical continuity, characterized in that, executed by a Logic Actor running in a local trusted isolation sandbox, the method comprises the following steps:
1. Logical causal sequence extraction: capture the demand feature vector, solution feature vector, physical action execution log hash generated by the local secure zone, and quantitative effect feedback feature vector in the current interaction session;
2. Causal self-consistency comparison: calculate the cosine similarity between the demand feature vector and the solution feature vector, and when the solution is invoked and the corresponding quantitative effect feedback feature vector is generated, determine the temporal self-consistency boundary using the physical action execution log hash as an index;
3. Data desensitization proof packaging: perform in-situ causal determination on the quantitative effect feedback feature vector and remove underlying sensitive plaintext including physical privacy parameters, to generate desensitized Boolean determination proof data containing the current interaction consistency score;
4. Hardware signature broadcast and transmission: perform cryptographic digital signature on the proof data using a hardware private key protected in the sandbox isolation zone, and broadcast the signature and the proof data packet to the distributed right confirmation ledger for on-chain sincerity synchronization.

Claim 9

The ledger contract-side operation method according to **Claim 7**, characterized in that:
The signature proof reception and verification step further includes an anti-replay verification sub-step: read the temporal random number or timestamp in the data packet, and compare it with the latest historical transaction identifier of the corresponding actor saved in the ledger; if the

difference between the two is outside the preset range or already exists in the historical records, determine it as an illegal replay request and reject it; and when updating the sliding window, add version concurrent state control based on a concurrent lock or optimistic lock to the update of the dynamic audit weight score.

Claim 10

The Logic Actor sandbox-side determination method according to **Claim 8**, characterized in that: In the hardware signature broadcast and transmission step, a one-time random **Nonce** of the current session is mixed in when performing digital signature with the hardware private key, so as to realize signature anti-replay; and before the signature and the proof data packet are sent to the distributed right confirmation ledger, physical transmission encryption is performed through an asymmetric public key encryption channel to prevent interception and snooping during transmission.

Claim 11

A computer-readable storage medium having a computer program stored thereon, characterized in that the program, when executed by a processor, implements the method according to any one of **Claim 4** to **Claim 10**.